Position, distance, displacement

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Distinguish a location, a total path length, and a change in position.

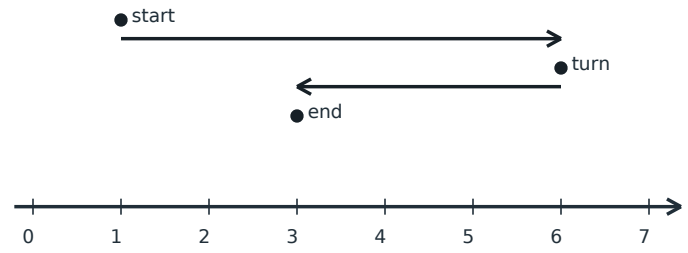
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 1.*
38 m if the 5 m/s rightward motion continues. Check a later timed position against the prediction.

02. Worked example: explain the model
Route arrows 4 m right and 2 m left; resultant 2 m right.

03. A walk along the hallway
Start 1 m, turn 6 m, end 3 m.

Model / diagram check



Position (m); right is positive

04. Count the whole path
5 m + 3 m = 8 m.

05. Connect start to finish
Displacement 2 m right; final position 3 m. They differ because the start was 1 m, not zero.

06. A complete trip back
Distance 10 m; displacement 0 m because start and end coincide.

07. Fix the wording
No. Distance is path length. Example: 4 m out and 4 m back gives 8 m distance and zero displacement.

08. Review: connect an earlier idea *Review: Unit 1, Day 1.*
Position increases by 5 m each second; equal rightward changes suggest steady motion.

09. Exit check
Distance 5+2=7 m; displacement 3 m right; final position 5 m.

10. STRETCH: test the model
Direct 1 to 4: 3 m. Route 1 to 6 to 4: 5+2=7 m. Both displacements 3 m right.

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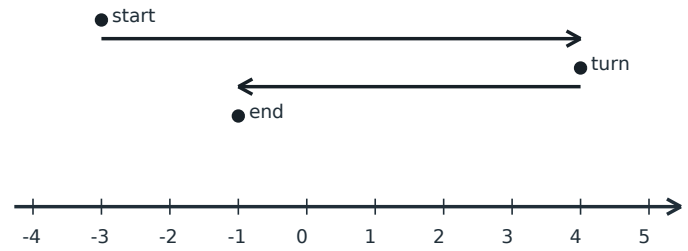
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 1.*
v=-2.5 m/s; x(7)=-2.5 m. Assumes constant velocity; compare another timed position with its uncertainty.

02. Worked example: explain the model
Route arrows 4 m right and 2 m left; resultant 2 m right.

03. Track a signed route
Distance 7+5=12 m; displacement -1-(-3)=+2 m.

Model / diagram check



Position (m); right is positive

04. Change the coordinate convention
Coordinates become +3,-4,+1 m; displacement -2 m. Physical route, distance 12 m, and displacement direction are unchanged.

05. A closed path
No. Distance is 500 m. The full lap returns to the start; the extra 100 m is a quarter-circle. Displacement is the chord, not the path. If calculated, r = 400/(2π) m and chord = r√2 ≈ 90.0 m. A labeled sketch and correct distinction are sufficient.

06. Different paths, same endpoints
Examples: east then north gives 7 m; direct diagonal gives 5 m. Both displacement (4, 3) m, magnitude 5 m.

07. Zero and nonzero
Yes: return trip. No: nonzero displacement requires a path of nonzero length; distance is at least displacement magnitude.

08. Review: connect an earlier idea *Review: Unit 1, Day 1.*
Motion is reported across 3 images. Uncalibrated images do not establish size, distance, full trajectory or impact; more measurements and uncertainties are needed.

09. Exit check
Distance 7+3=10 m; displacement 1-5=-4 m; final position +1 m. They describe path length, change, and location respectively.

10. STRETCH: test the model
Turn at +6 m: 8+3=11 m; or -5 m: 3+8=11 m. Displacement +5 m in either case.